MILITARY SPECIFICATION

CLOTH, MELTON, WOOL (MOTH-PROOFED)

This specification is approved for use by Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for wool melton cloth.

1.2 Classification The melton cloth shall be of the following types and classes (see 6.2):

Type I - 16 ounce
  Class 2 - Blue 3346

Type II - 22 ounce
  Class 4 - Blue 3346

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the soliciation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-C-43665 - Cloth, Wool: Mothproofing Treatment of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool, and Wool Blends

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE WOOL PRODUCTS

LABELING ACT OF 1939

(Application for copies should be addressed to the Superintendant of Documents, Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE

Methods of Test for Grades of Wool

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish sample cloth for first article inspection and approval (see 4.2 and 6.2).

3.3 Material.

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3.3.1 Wool stock. The wool stock shall be of fleece or pulled wool, or a combination of both, not lower in grade than 64's U.S. Standard (see 4.3.1.1). The use of noils, laps or any other wool manufacturing by-products is prohibited.

3.3.2 Yarns. The wool stock shall be thoroughly scoured, blended, carded and evenly spun on the woolen system into singles yarn for both warp and filling.

NOTE: Selvage yarns - To prevent edges of the cloth from building up when rolled, the single yarn used in the selvage may be made of a finer count than those used in the body of the cloth. These yarns shall also consist of wool fibers.

3.4 Weave. The weave for type I shall be a 2 up, 1 down right hand twill and for type II, the weave shall be a 3 up, 1 down crowfoot when tested as specified in 4.4.

3.5 Physical requirements. The finished cloth shall conform to the requirements listed in Table I when tested as specified in 4.4.

Table I - Physical requirements

	Weight ozs (grams) linear yard 1/		Texture yarns per inch (2.54cm) Minimum		Breaking strength (pounds) Minimum		Shrinkage (percent) Maximum	
Type	Min.	Max.	Warp	Filling	Warp	Filling	Warp	Filling
I	16 (448g)	17 (476g)	60	55	58	46	4.0	3.0
II	22 (616g)	24 (672g)	55	45	80	60	4.5	2.0

1/ Based on 56 inch width.

*3.5.1 Width. The width of all types and classes of the finished cloth shall be a minimum of 56 inches exclusive of selvages or tuck in when woven on sulzer type looms.

3.6 Wool content. The finished cloth shall contain not less than 95.0 percent wool when tested as specified in 4.4.

3.7 Color. The color of the finished cloth shall be shade Blue 3346.

3.7.1 Dyeing. The color shall be produced by stock dyeing with suitable dyestuffs of required fastness properties to match the standard sample.

3.7.2 Matching. The color of the finished cloth shall match the standard sample under artificial daylight, having a color temperature of 7500 degrees Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800 degrees Kelvin.

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*3.7.3 Colorfastness. The finished cloth shall show colorfastness to wet-dry cleaning, light, laundering, perspiration (acid test only) and crocking equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is available, the dyed cloths shall show good fastness to wet-dry cleaning, light, laundering, perspiration (acid test only) and shall show a Munsell Value for crocking not lower than 8.5 when tested as specified in 4.4.

3.8 Finish. The type and character of the finish shall be equal to or better than the standard sample. The cloth shall be thoroughly scoured, properly fulled, (carbonized if necessary) face evenly sheared and well pressed.

3.9 Length and put-up. Unless otherwise specified (see 5.1.1 and 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (36.56m). Each length shall be put up in full width rolls as specified in PPP-P-1132.

3.10 Wool content label. Cloth manufactured under this specification shall be labeled in accordance with the Wool Products Labeling Act of 1939.

3.11 Mothproofing. The cloth shall be given a mothproofing treatment in accordance with MIL-C-43665 (see 6.2).

3.12 pH. The pH of the water extract of the finished cloth shall be no less than 4.0 nor more than 8.0 when tested as specified in 4.4.

3.13 Marking. The back of the cloth shall be marked "Back" at both ends of each roll with letters not less than 1/2 inch (12.7mm) in height. The marking shall be clearly legible in any indelible marking used commercially and shall not show through on the face of the cloth.

3.14 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.61m^2$) when calculated as specified in Section 4, shall not exceed the established maximum point value. Each defect shall be marked with a 1 to 1 1/2 inch (2.54cm to 3.81cm) long string. The string shall be sewn into the selvage near the defect. A red string shall represent a three or four point defect and a white string shall represent a one or two point defect (see 4.3.2.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform in prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test items to determine the validity of the certification.

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4.2 First article inspection. When required, the first article of the finished cloth submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical properties in accordance with the methods specified in 4.4.

4.3 Inspection. Inspection shall be performed in accordance with MIL-STD-655, except where otherwise indicated.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Wool. The wool shall be visually examined for grade in scoured form, prior to dyeing, picking or carding by comparison with the applicable US Standard (see 2.2). The examination for grade shall be performed on a composite sample of 1 pound (0.45Kg) of scoured wool for each 10,000 pounds (4500Kg) or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bags/bales in the lot. In the event of dispute resulting from above comparisons, the grade shall be determined by a width method (wedge). The lot shall be unacceptable if one or more composite sample fails to meet the requirements of 3.3.1. Each lot shall consist of one grade of wool only.

4.3.1.2 In-process inspection. Inspection shall be made at any point during any phase of the manufacturing process to assure that no laps, noils, or any other wool manufacturing by-products are used (see 3.3.1). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance.

4.3.2 Examination of the end item. Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.2.5.

4.3.2.1 Yard-by-yard examination. The yard-by-yard examination shall be as specified in MIL-STD-655. The fabric quality for the finished cloth shall be quality level 10.

4.3.2.1.1 Limit of slubs and knots. Only slubs and knots which exceed "D" for slubs and "C" for knots on the Sears Fabric Defect Scale (see 6.4) shall be scored.

4.3.2.2 Examination for length. The examination for length shall be as specified in MIL-STD-655.

4.3.2.3 Examination for back marking and compliance with the Wool Products Labeling Act. The examination for back marking and compliance with the Wool Products Labeling Act shall be as specified in MIL-STD-655.

4.3.2.4 Examination for shade match and finish, individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.7.2), and finish (see 3.8). A roll shall be unacceptable if it fails to meet the requirements for shade match or finish as applicable. The sample unit shall be a 4 inch by 20 inch (10.1 by 50.8cm) swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

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4.3.2.5 Examination for defect marking (stringing of defects). The examination for defect marking shall be as specified in MIL-STD-655.

4.3.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1132 to determine that packaging, packing and marking comply with the Section 5 requirements of this specification.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II, shall be followed. The physical and chemical values specified in Section 3, except where otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3 1/2 yards (3.19m) full width of the finished cloth. The lot size shall be expressed in units of one yard (.914m). The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Test methods

Characteristic	Requirement paragraph	Test method
Weave	3.4	Visual 1/
Weight	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5100
Shrinkage	Table I	5590
Colorfastness to:		
Laundering	3.7.3	5614 2/
Wet-dry cleaning	3.7.3	5622
Crocking	3.7.3	5651
Light	3.7.3	5660
Perspiration 3/	3.7.3	5680
Wool content	3.6	2101
Mothproofing content	3.11	4/
pH	3.12	2811

1/ One determination shall be made and the results reported as "Pass" or "Fail".
2/ Only the stain on the wool fibers of the color transfer cloth shall be evaluated.
3/ For acid prespiration test only.
4/ As specified in MIL-C-43665.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be packaged in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with applicable requirements of PPP-P-1132.

5.3 Marking. In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in clothing worn by Naval personnel.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type and class of cloth required (see 1.2).
c. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.
d. Minimum and maximum lengths for cloth (see 3.9).
e. Moth repellency requirement (see 3.11).
f. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids (see 3.1).

6.4 Sears Fabric Defect Scales are avilable from Sears, Roebuck and Co., Department 733Q. "Sears Defect Replica Kit", Sears Tower, Chicago, ILL 60684.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evalute the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-16290J(NU)	CLOTH, MELTON, WOOL (MOTH-PROOFED)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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